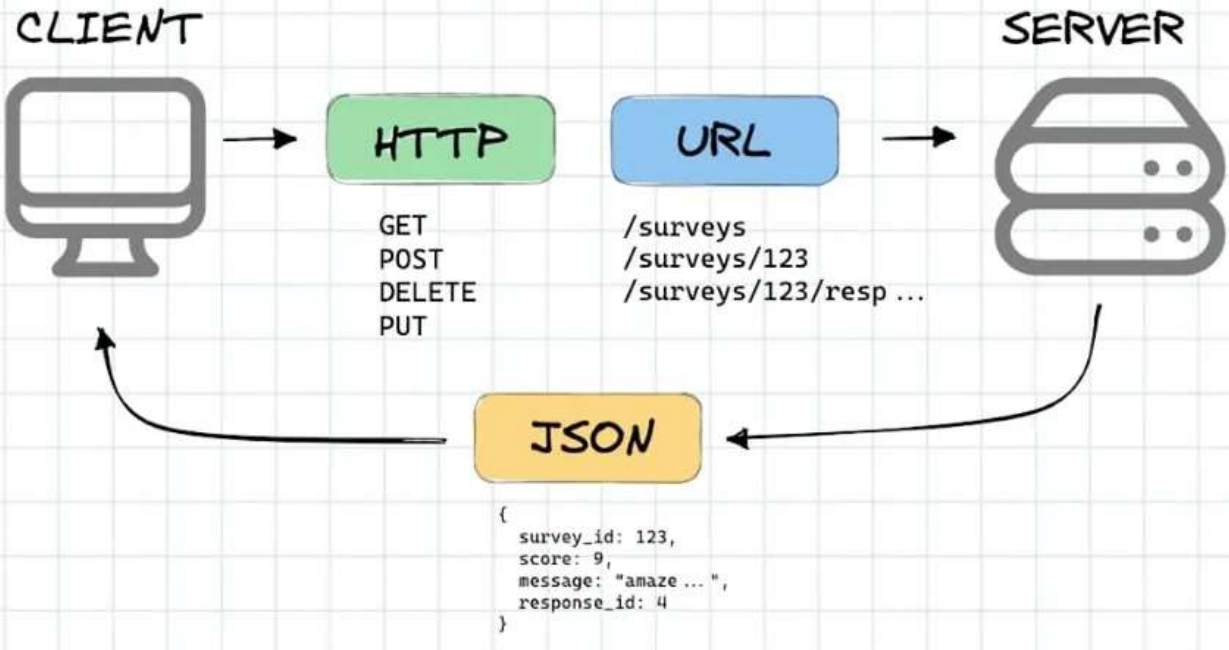
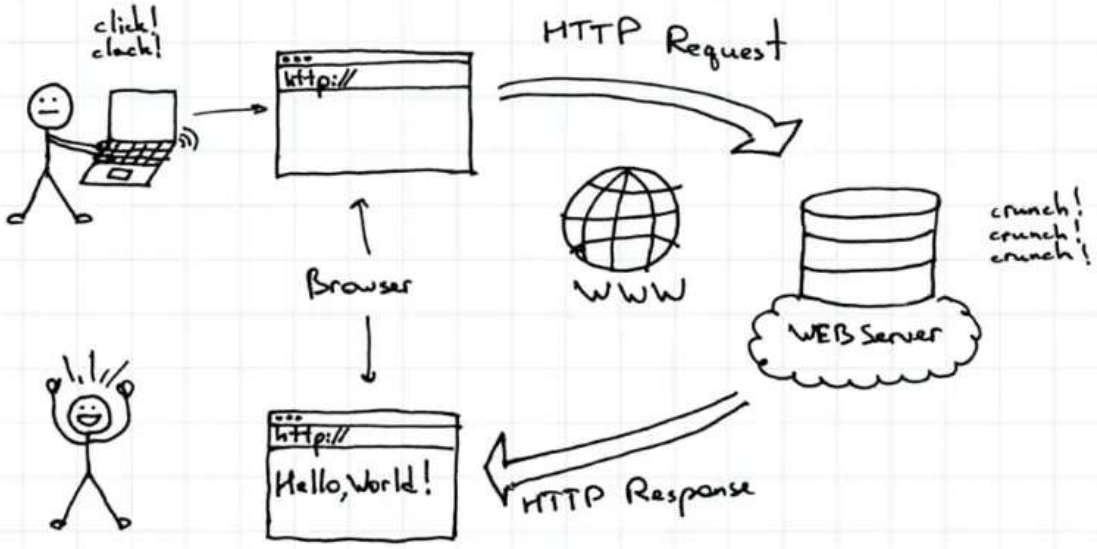
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REST API EXPLAINED



What is a REST API?

A REST API (**Representational State Transfer Application Programming Interface**) is a set of rules that lets applications communicate over HTTP. It follows the REST principles, an architectural style defined by Roy Fielding.



Key Principles of REST

- ***Stateless:*** Each request must contain all necessary information, as the server doesn't store client session data.
- ***Client-Server Architecture:*** The client and server work independently, ensuring a clear separation of concerns.
- ***Cacheable:*** Responses can be marked as cacheable to improve performance.
- ***Uniform Interface:*** REST APIs follow standard HTTP methods (GET, POST, PUT, DELETE) and consistent resource conventions.
- ***Layered System:*** The architecture supports multiple layers, improving scalability and security.
- ***Code on Demand (Optional):*** Servers can send code (e.g., JavaScript) to extend client functionality.

REST API operations

HTTP Method	CRUD Operation	Description
GET	Read	Retrieve data from the server.
POST	Create	Send data to the server to create a resource.
PUT	Update	Update an existing resource.
DELETE	Delete	Remove a resource.
PATCH	Update (Partial)	Partially update a resource.

REST API Components

Component	Description	Example
Resources	The entities or data the API interacts with, represented as URLs.	/users, /products, /orders
Endpoints	The specific URLs where the API can be accessed.	https://api.example.com/v1/users
HTTP Status Codes	Standard codes indicating the outcome of an API request.	See below

Common HTTP Status Codes

Status Code	Meaning
200 OK	Request succeeded.
201 Created	Resource created successfully.
400 Bad Request	Invalid input or request.
401 Unauthorized	Authentication required.
404 Not Found	Resource not found.
500 Internal Server Error	Server-side error.